This Step or From Here On? Crossed Same-State Interventions for Evaluating Quantized Agents

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Abstract

When a quantized agent and its BF16 reference reach similar average task success, but the quantized agent takes a different action at some state, should BF16 take over, and if so for this one step or from here on? We argue that the answer depends on the state and on the controller that continues, so average success, action divergence and the gain from precision replacement must be measured separately. We propose a crossed same-state intervention: at a frozen, bit-exactly replayable state, we cross the source of the current action (BF16 or quantized) with the controller of the remaining episode. This yields three replacement gains on the same states: the current action only (G), the continuation only (H), or both (F). On Qwen3-4B and Qwen3-8B in τ^2-bench (85 tasks, 20 nonce blocks), replacing a W4 agent's first divergent action changes the task outcome in 15.5–20.9% of task-block units, in both directions, while the net gain is between -1.9 and $+2.9$ pp. In about a quarter of the divergent states that each W4 cell samples (pooled 26.9%; new-task interval 22.1–32.0%), the outcome effect of the same action swap changes with the controller that continues. Strict sign reversals are only 1.2–4.2% of states in the W4 and KV8 cells, and dependence in the two directions is about equally frequent, so the mean interaction is near zero. On the fixed task pool, replacing the current action gains more than replacing the continuation in both q8 W4 cells, and less in a severe W3 stress cell; these contrasts are within cells, and their intervals for new tasks cover zero in the W4 cells. At outcome-independent states, most outcome-changing differences under our user simulator are text differences with the same tool structure. These outcome labels did not reproduce reliably under an alternative, stochastic relay simulator. Quantized-model uncertainty marks where actions diverge; a frozen uncertainty trigger enriches outcome-changing decisions on later blocks but has not been shown to select net-beneficial replacements. On 34 unseen telecom tasks under a pre-registered protocol, the action swap again both rescues and damages, task fragility repeats across independent nonce halves, and the average gain is -4.6 pp (90% CI -7.2 to -2.0), which the protocol's pre-specified readings classify as inconclusive.

1 Introduction

Quantized LLM agents are usually evaluated by their average task success next to that of the full-precision reference. In a closed-loop benchmark such as τ^2-bench (Barres et al., 2025), that comparison can look reassuring. On our checkpoints, quantized-minus-BF16 success ranges from -0.2 to $+4.2$ pp across four W4 cells. Its intervals are about $\pm$6–7 pp wide, so an equivalence test at ±3 pp fails everywhere (Appendix B). Yet the quantized agent (Q) still takes actions that its BF16 reference (B) would not. If B is available as a fallback, as in dynamic precision routing (Li et al., 2026c), a practical question arises at every such state: **is it worth letting BF16 take over, and should it take over just this step or from here on?**

Judging the two actions does not answer this. Consider a state from our data (q4 AWQ-W4, τ^2-bench airline task 31, nonce block f13, decision 1 of 12). The user wants to move an upcoming flight. Q looks up a reservation id that does not exist and receives "not found"; B looks up the user's profile. B's action plainly looks better. We froze this state and ran all four combinations of {B's or Q's action} $\times$ {B or Q continuing}. Only one succeeds: B's action *followed by* B. If B supplies only this step and Q continues, Q searches for flights, writes a change and fails. If Q's failed lookup stands and B continues, B also fails. The better-looking action has value only under one continuation; elsewhere, a handover to B after Q's error

damages a task that Q would have recovered by itself (§8). Whether a local action difference has task consequences depends on the state and on the subsequent controller. **Average success, action divergence and precision-replacement gain must therefore be measured separately.**

What we do. We propose an interventional evaluation methodology for quantized agents and use it for a systematic empirical study. At a frozen state whose no-op replay is bit-exact, we cross the source of the current action with the controller of the remaining episode (Figure 1). Three paired gains answer the three escalation questions on the same states. G replaces only the current action, H replaces only the continuation, and F replaces both (§3). We apply the design to two state banks: first divergences between B and Q along Q's trajectory, and ordinary states drawn independently of any outcome.

What we find.

- **R1, replacing the current action** (§4). Replacing a W4 agent's first divergent action changes the outcome in 15.5–20.9% of task-block units, rescuing about as often as damaging; the net gain is -1.9 to $+2.9$ pp. This is a local effect, not end-to-end equivalence.
- **R2, where to replace** (§5). In about a quarter of W4 divergent states (pooled 26.9%), the action swap's effect on the outcome depends on who continues, although strict sign reversals are rare and the mean interaction is near zero. Within cells the paired gains can be compared directly: on the fixed task pool, replacing the current action gains more than replacing the continuation in both q8 W4 cells and less in a severe W3 stress cell; orderings across cells are descriptive.
- **R3, sampling and environment** (§6). On outcome-independent states, most outcome-changing differences under our simulator are text differences with the same tool structure. These outcome labels did not reproduce reliably under an alternative, stochastic relay simulator. The two-sided pattern of R1 also appears on BFCL and ALFWorld, which have no user simulator.
- **R4, finding replacements in advance** (§7). Q's token-level uncertainty locates where B and Q diverge, but not which divergences matter.

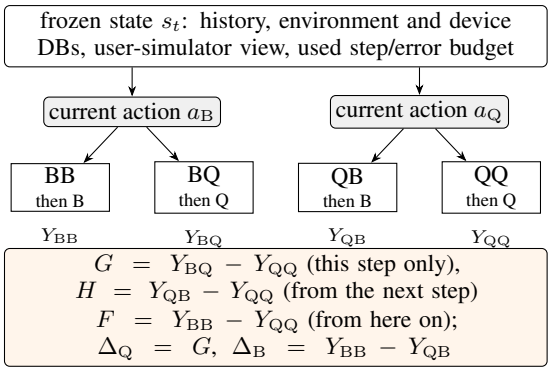


Figure 1: Crossed same-state intervention (schematic, not data). B = BF16 agent, Q = quantized agent. In branch XY, the first letter is the source of the current action and the second is the controller from the next step to the end. Every branch starts from the same frozen state with the original remaining budget. QQ re-executes Q's original episode, and its bit-exact reproduction is the no-op check. Definitions are in §3.2.

Contributions. (1) A crossed same-state intervention for quantized agents, with paired estimands (G, H, F and the (Δ_Q, Δ_B) table), bit-exact replay acceptance, explicit denominators, and inference that separates a fixed task pool from new tasks. (2) A systematic study over weight, KV-cache and stress quantization, two model sizes and three benchmarks. (3) Evidence that the consequences of a quantization-induced action difference are state- and controller-dependent and two-sided, so precision replacement can help or hurt. Withdrawn results are in Appendix I.

2 Related Work

We position prior work by the quantity it identifies.

Episode-level comparisons of quantized agents. *Flat Score, Amplified Failures* (Jang et al., 2026) is the closest study in setting. On τ^2-bench retail and telecom it shows that 4-bit agents keep a flat task score while process damage grows. It reports a two-sided paired task split (33 tasks worse, 26 better for Gemma-4-31B on telecom). It explains the flat score by the error budget, through a counterfactual projection from logs, and by recovery, which is equally frequent at both precisions (55% after a failed call). It also tests a repair prompt. Its comparisons are between independently sampled episodes, so the precision of the action that went wrong and the precision of the controller that must recover change together. Our design holds the state fixed and varies the two separately. H measures what the recovering controller contributes on

identical histories, and the (Δ_Q, Δ_B) table measures whether an action's consequence depends on that controller. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); agent-compression benchmarks report averages (Dong et al., 2025).

Live forks with a model switch. *The Replay Gap* (Gonuguntla, 2026) forks live SWE-agent trajectories at a fixed step and continues them with a different model (Qwen3-4B FP8 vs. Qwen3-14B AWQ), against same-model control forks. At the switch point the new model produces the next action *and* the continuation, so the two change together; this is our F contrast, not G or H. Its 359 control forks had 0 success flips, and its FP8-served controls diverged in action sequence on 90–96% of forks. We claim no noise advantage over that work. Our zero replay noise is a property of our deterministic serving (§3.3). Single-step action swaps under a fixed continuation fill one cell of our design and are used for training (Li et al., 2026a,b), attribution (Shah, 2026) and locating decisive errors (Zhang et al., 2025). None of the 32 closest works we checked crosses the action with the continuation controller at a precision-induced divergence.

Precision routing and its proxies. DMR (Li et al., 2026c) learns a per-step BF16/W4 router from BF16-versus-quantized KL supervision and episode reward. Precision sensitivity has also been localised to phases (Lu et al., 2026) and task types (Zhang et al., 2026). We do not propose a competing router. We measure what a router would harvest, namely the intervention gains G, H and F at given states, and we test whether a Q-visible proxy predicts them. For quantized reasoning models, Lotfi et al. (2026) trace damage to high-entropy, high-KL token positions, where the quantized model over-samples markers such as "wait". A logit penalty on those markers both repairs the damage and validates the mechanism. §7 tests the analogous uncertainty view for agent actions.

Evaluation noise and user simulators. Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit, and runs were bit-identical with the cache off (Patodiya, 2026). Agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026). Changing the user model moves τ-bench retail success by up to 9 points (Seshadri et al., 2026) and reward by 15.2 points (Jain and Sandhu, 2026).

3 Crossed Same-State Intervention

3.1 Setting

Tasks and agents. We use τ^2-bench in text mode. The development panel has 58 retail and 27 airline tasks plus 80 telecom development tasks. A further 34 telecom tasks form a holdout that no agent ran before its protocol was frozen. The agents are Qwen3-4B-Instruct (q4) and Qwen3-8B with thinking disabled (q8). The user simulator is one greedy Qwen3-30B-A3B-Instruct model shared by all agents. Retail and airline success is the final-database match; telecom uses the official task-basis success. Without a user simulator, we also run BFCL multi-turn and ALFWorld with one run per item (Appendix A).

Quantization and serving. Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B. The main cells are GPTQ (Frantar et al., 2023) and AWQ (Lin et al., 2024) at W4 and a KV8 cache; stress cells are GPTQ-W3, RTN-W3 and a Q/K-rotated KV4 cache. Because quantization is simulated, we make no speed or memory claim. B and Q run on vLLM with batch-invariant kernels and deterministic compilation, in separate server processes, so no prefix cache is shared across weights. A *block* is one closed-loop run of Q over the task pool under one prompt nonce, a random session-identifier line in the system prompt. Independent nonces give independent blocks, the replicate unit.

3.2 Frozen states and estimands

Frozen state. Before Q's decision t in a recorded trajectory, we restore the state s_t. It comprises the full message history; the environment and user-device databases, rebuilt by strict replay of every recorded tool call; the user simulator's visible conversation; and the used step and error budget. B and Q each generate a full action from the same request, and the prompt token ids both servers receive must be identical. The state is written to disk before any branch runs.

Crossed intervention. $Y_{XY} \in \{0, 1\}$ is task success when the action at s_t comes from X and controller Y takes over from the next step to the end, with X, Y $\in \{B, Q\}$. The action is executed by the

real environment or answered by the real simulator and consumes one step; all four branches start from the same frozen state with the original remaining budget. QQ re-executes Q's own episode, and BB hands both the step and the rest of the episode to B. Each escalation option is a paired contrast with QQ:

$$G = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}} \quad \text{(this step only)},$$
$$H = Y_{\mathrm{QB}} - Y_{\mathrm{QQ}} \quad \text{(from the next step)},$$
$$F = Y_{\mathrm{BB}} - Y_{\mathrm{QQ}} \quad \text{(from here on)}.$$

Their paired differences compare the options directly: $G-H = Y_{\mathrm{BQ}}-Y_{\mathrm{QB}}$, $F-G = Y_{\mathrm{BB}}-Y_{\mathrm{BQ}}$ and $F - H = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$. The effect of swapping the action can be read under either controller: $\Delta_{\mathrm{Q}} = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}}$ (Q continues; equal to G) and $\Delta_{\mathrm{B}} = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$ (B continues). Each takes values in $\{-1, 0, +1\}$, which gives a 3×3 table per state population. On its diagonal, the action swap has the same effect under both controllers. Off the diagonal, the interaction $I = \Delta_{\mathrm{Q}} - \Delta_{\mathrm{B}}$ is nonzero. The two corners $(+1, -1)$ and $(-1, +1)$ are *strict reversals*: the better action under one controller is the worse one under the other. These contrasts judge consequences, not correctness.

Rescue and damage. For the current-action swap, a *rescue* is $(Y_{\mathrm{BQ}}, Y_{\mathrm{QQ}}) = (1, 0)$ and a *damage* is $(0, 1)$. With rates r and d, the outcome turnover is $M = r + d$ and the net gain is $r - d$.

3.3 Replay acceptance

Causal reading requires exact replay. Before any development outcome was analysed, an engineering gate on real divergence states in retail, airline and telecom passed every check. Q→Q no-ops matched in 20/20 states and B→B no-ops in 10/10, in every prompt-token hash, message, database hash, termination, reward and budget counter. In production, a sampled QQ no-op passed 924/924 units of the development analysis, and the outcome-independent bank passed 135/135 no-ops. The replay noise of a branch is therefore zero. This follows from our serving, which closes the prefix-cache channel (Patodiya, 2026); we do not claim it as an advantage over prior forks.

3.4 State banks

Bank P: first tool-structure divergences. For each task, block and quantized cell, we walk along Q's trajectory to the first decision at which a_{B} and a_{Q} differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments. **Differences in text only are skipped**: two different natural-language replies, or identical calls with different accompanying text, are not Bank P states, and the walk continues past them. A task-block unit without such a divergence has no Bank P state. Bank P is selected by construction: it is the first point at which tool structure differs.

Bank U: ordinary states. Bank U draws one decision per task uniformly, with the draw fixed by a hash, and its manifest was frozen before any generation. It includes *all* full language actions: every state runs all four branches unless a_{B} and a_{Q} are token-identical, and no effect is imputed. U1 states come from each quantized cell's own trajectories. U2B states come from BF16 trajectories and U2Q states from Q trajectories; every cell of a model receives the identical U2 state, which allows same-state comparison across quantizers. A frozen 25% subsample of U2 is also continued by one common controller Q0, the GPTQ-W4 checkpoint of the same model.

3.5 Denominators and inference

We use three denominators and never mix them. (1) *Full pool*: every task × block unit of a cell, with units that have no Bank P state counted as zero. r, d, M and the net gain $G_{\mathrm{all}} = r - d$ use it (§4). (2) *Divergent states*: the Bank P states of one cell, as a ratio of block sums. G, H, F and the 3×3 table use it (§5); G_{all} is G diluted by the share of units without divergence. (3) *Bank U*: one state per task, with equal task weight, reported per sub-bank (§6).

All tasks of a block share its nonce, and all branches share each state. Every within-cell contrast is therefore paired and *identified* for that cell's state population. Comparisons across cells (q4 vs. q8, AWQ vs. GPTQ, W4 vs. KV8) compare different state populations, because each cell selects its own first divergences, so they are *descriptive*. Only complete blocks enter; a block with any missing unit leaves the analysis. The primary interval treats the task pool as fixed: a block bootstrap with at least 10 complete blocks, otherwise a t interval over block means (stress cells have 3 blocks). A crossed task × block bootstrap is the interval for generalising to new tasks. Bank U is analysed over blocks f01–f03; its primary intervals are t intervals

Cell	units	div.	r	d	M	G_{all} [fixed pool]	[new tasks]
q4 GPTQ-W4	85×20	1102	6.8	8.7	15.5	-1.9 [-3.7, -0.1]	[-6.8, $+2.7$]
q4 AWQ-W4	85×20	1439	9.8	10.1	19.9	-0.3 [-2.2, $+1.6$]	[-5.4, $+5.2$]
q8 GPTQ-W4	85×20	1473	10.4	7.5	17.9	$+2.9$ [$+1.4$, $+4.4$]	[-2.9, $+9.2$]
q8 AWQ-W4	85×20	1562	11.4	9.5	20.9	$+1.9$ [-0.6, $+4.3$]	[-3.9, $+8.1$]
W4 pool	85×20		9.6	9.0	18.6	$+0.7$ [-0.3, $+1.6$]	
q4 KV8	85×22	593	3.1	3.4	6.5	-0.3 [-1.7, $+1.0$]	[-3.0, $+2.3$]
q8 KV8	85×22	855	4.3	3.7	8.1	$+0.6$ [-0.4, $+1.6$]	[-1.9, $+3.0$]
q4 GPTQ-W3^s	85×3	232	9.0	10.2	19.2	-1.2 [-7.0, $+4.7$]	[-8.6, $+5.9$]
q4 KV4-rot^s	85×3	233	11.8	7.1	18.8	$+4.7$ [-4.1, $+13.5$]	[-3.1, $+12.9$]
q8 GPTQ-W3^s	85×3	242	10.2	8.2	18.4	$+2.0$ [-12.5, $+16.4$]	[-6.3, $+11.8$]
q8 KV4-rot^s	85×3	239	11.8	8.6	20.4	$+3.1$ [-14.5, $+20.8$]	[-5.9, $+12.9$]
q8 RTN-W3^s	85×3	255	3.1	3.5	6.7	-0.4 [-3.8, $+3.0$]	[-4.7, $+3.9$]

Table 1: Replacing Q's first divergent action while Q continues, τ^2-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix B). Fixed pool: block bootstrap (≥10 blocks) or t over block means; new tasks: crossed task×block bootstrap. ^{s}stress cell, 3 blocks. Source: `analysis/fork/out/dev_all_20261001T070326Z.md`.

over the three block means.

4 R1: What Replacing the Current Action Rescues and Damages

Two-sided turnover, small net gain. Table 1 reports the current-action swap with its denominators. On the τ^2-bench panel (85 tasks × 20 complete blocks), replacing a W4 agent's first divergent action changes the task outcome in $M = 15.5$–20.9% of units. Rescues and damages are of similar size, $r = 6.8$–11.4 and $d = 7.5$–10.1 pp, so the net gain G_{all} lies between -1.9 and $+2.9$ pp. Two fixed-pool intervals exclude zero, negative for q4 GPTQ-W4 and positive for q8 GPTQ-W4; none of the new-task intervals does. Pooled over the four W4 cells, G_{all} is $+0.7$ [-0.3, $+1.6$] pp with $M = 18.6\%$. KV8 turns over fewer outcomes ($M = 6.5$–8.1%), and the stress cells reach W4-like turnover with only three blocks. A small net gain therefore coexists with substantial turnover in both directions.

What this is not. G_{all} is the effect of replacing one action at a selected state while Q continues. It says nothing about whether the quantized and BF16 agents are equally successful end to end. That comparison runs two independent trajectories, and on the same checkpoints, tasks and blocks it is inconclusive: an equivalence test at ±3 pp fails in every W4 cell (Appendix B).

Turnover concentrates in tasks (exploratory). The top 20% of tasks hold 57–71% of all flips in the W4 and KV8 cells, and task-level turnover and its direction repeat across disjoint halves of the nonce blocks. This is a retrospective association on viewed development data (Appendix F); its pre-registered test on unseen tasks is pending (§6).

5 R2: Where to Replace

The three escalation options of the introduction are G (this step), H (from the next step) and F (from here on). Table 2 reports them on the same states of each cell, with the paired differences that compare them directly.

On average over W4 cells, no option has a detectable gain. Pooled over the four W4 cells, $G = +0.5$, $H = +0.2$ and $F = +0.7$ pp, and every interval covers zero, as do the differences between them. The pool hides cell-level structure, which we now read within cells, where it is identified.

Within cells, the paired gains differ. In q8 GPTQ-W4, replacing the current action gains $G = +3.3$ [$+1.6$, $+5.0$] pp, replacing only the continuation gains $H = +0.4$ [-1.2, $+1.9$], and the difference is $G - H = +2.9$ [$+1.3$, $+4.7$]. Replacing both gains $F = +5.4$ [$+3.5$, $+7.2$], with $F - G = +2.1$ [$+0.2$, $+4.0$]. In q8 AWQ-W4, $G - H = +3.4$ [$+1.5$, $+5.2$] and $F - G = -1.5$ [-4.2, $+0.8$]. In q4 GPTQ-W4, $G = -2.9$ [-5.7, -0.2] and $G - H = -3.6$ [-6.6, -0.6]. In the stress cell q8 RTN-W3 (three blocks, exploratory), $G = -0.4$ [-3.8, $+3.0$], $H = +15.3$ [$+0.7$, $+29.9$], $F = +18.4$ [$+16.7$, $+20.1$] and $G - H = -15.7$ [-27.5, -3.9]. In q4 KV8, B continuing after Q's action lowers success: $H = -2.5$ [-4.5, -0.5]. These intervals treat the 85 tasks as fixed. Holm-adjusted over the four core W4 cells, $G - H$ remains below .05 in both q8 cells ($p = .013, .012$) but not in q4 GPTQ-W4 ($p = .069$), and for new tasks the $G - H$ intervals of all four W4 cells cover zero (Table 5). We therefore treat the $G - H$ results as exploratory. Each statement compares contrasts within one cell. Cells select different states, so an ordering across cells is descriptive and cannot be attributed to the quantizer or the bit width alone.

A BF16 reference action can lower success. In q4 GPTQ-W4, replacing Q's action with B's lowers success under Q's continuation ($G = -2.9$ [-5.7, -0.2] pp) and, by a similar amount, under

Cell	states	G (this step)	H (next step on)	F (from here on)	$G - H$	$F - G$	$F - H\ (= \Delta_B)$	off-diag. %	rev. %	mean I
q4 GPTQ-W4	1102	−2.9 [−5.7, −0.2]	+0.7 [−1.6, +3.0]	−2.5 [−5.3, +0.2]	−3.6 [−6.6, −0.6]	+0.5 [−1.2, +2.2]	−3.2 [−6.4, +0.2]	25.0	2.6	+0.3
q4 AWQ-W4	1439	−0.3 [−2.6, +1.9]	+0.9 [−1.1, +2.9]	−0.8 [−3.3, +1.6]	−1.3 [−3.0, +0.6]	−0.4 [−2.4, +1.5]	−1.7 [−3.6, +0.4]	29.3	3.1	+1.3
q8 GPTQ-W4	1473	+3.3 [+1.6, +5.0]	+0.4 [−1.2, +1.9]	+5.4 [+3.5, +7.2]	+2.9 [+1.3, +4.7]	+2.1 [+0.2, +4.0]	+5.0 [+3.2, +6.8]	22.9	2.7	−1.7
q8 AWQ-W4	1562	+2.1 [−0.6, +4.7]	−1.3 [−3.5, +0.9]	+0.6 [−1.5, +2.7]	+3.4 [+1.5, +5.2]	−1.5 [−4.2, +0.8]	+1.9 [−0.1, +3.8]	30.5	4.2	+0.3
W4 pool	5576	+0.5 [−0.7, +1.7]	+0.2 [−1.3, +1.6]	+0.7 [−0.6, +2.1]	+0.4 [−0.8, +1.7]	+0.2 [−1.0, +1.2]	+0.5 [−1.0, +2.1]	26.9	3.2	+0.0
q4 KV8	593	−1.0 [−5.3, +3.2]	−2.5 [−4.5, −0.5]	−1.2 [−4.7, +2.3]	+1.5 [−3.1, +6.0]	−0.2 [−3.5, +3.7]	+1.3 [−2.5, +4.9]	15.7	1.2	−2.4
q8 KV8	855	+1.3 [−0.9, +3.4]	−0.4 [−2.1, +1.5]	+2.5 [+0.0, +5.1]	+1.6 [−0.6, +3.9]	+1.2 [−0.5, +3.2]	+2.8 [+0.3, +5.5]	14.9	1.2	−1.5
q4 GPTQ-W3^s	232	−1.3 [−7.7, +5.0]	+4.3 [−10.7, +19.4]	+1.3 [−1.9, +4.4]	−5.6 [−15.0, +3.7]	+2.6 [−0.6, +5.8]	−3.0 [−15.1, +9.1]	31.5	3.9	+1.7
q4 KV4-rot^s	233	+5.2 [−4.5, +14.9]	+9.4 [+5.1, +13.7]	+12.4 [+11.0, +13.9]	−4.3 [−18.3, +9.8]	+7.3 [−2.0, +16.5]	+3.0 [−1.9, +8.0]	33.0	1.7	+2.1
q8 GPTQ-W3^s	242	+2.1 [−13.3, +17.2]	+0.4 [−7.4, +8.2]	+2.1 [−7.4, +11.4]	+1.7 [−9.1, +12.3]	+0.0 [−11.0, +11.1]	+1.7 [−0.1, +3.4]	23.1	4.5	+0.4
q8 KV4-rot^s	239	+3.3 [−15.4, +21.9]	−0.8 [−18.2, +16.3]	+4.6 [−16.0, +25.0]	+4.2 [+2.5, +5.8]	+1.3 [−1.8, +4.3]	+5.4 [+0.8, +10.0]	31.0	4.6	−2.1
q8 RTN-W3^s	255	−0.4 [−3.8, +3.0]	+15.3 [+0.7, +29.9]	+18.4 [+16.7, +20.1]	−15.7 [−27.5, −3.9]	+18.8 [+13.8, +23.9]	+3.1 [−13.0, +19.2]	29.0	0.4	−3.5

Table 2: Three replacement gains and their paired differences on each cell's Bank P states (pp; same denominator in every column). $G = Y_{BQ} - Y_{QQ}$, $H = Y_{QB} - Y_{QQ}$, $F = Y_{BB} - Y_{QQ}$. Off-diag.: share of states with $\Delta_Q \neq \Delta_B$; rev.: strict reversals. Fixed-pool 95% intervals (block bootstrap over 20–22 complete blocks; t over 3 blocks for ^{s}stress cells, which cannot be pooled). New-task intervals are in Table 5. Within a row, contrasts are paired and identified; between rows, descriptive. Source: `analysis/fork/out/ghf_20261001T211519Z.md`.

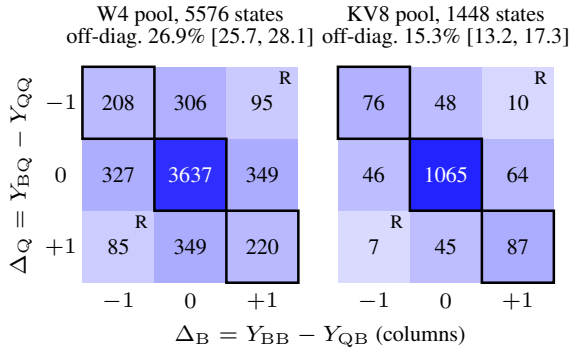


Figure 2: Does the action swap have the same effect under both controllers? Counts of (Δ_Q, Δ_B) over Bank P states, summed over the W4 cells (left) and KV8 cells (right). Boxed: diagonal. R: strict reversals. Shading: share of states (power scale). Off-diagonal shares: equal-weight cell averages, fixed-pool 95% intervals. Source: `analysis/fork/out/ghf_20261001T211519Z.md` (§2).

B's ($\Delta_B = -3.2$ [−6.4, +0.2]). We do not attribute this to the continuing controller: the reference action itself is not reliably better here.

The action's consequence often depends on who continues. Figure 2 shows the (Δ_Q, Δ_B) table. In 26.9% [25.7, 28.1] of pooled W4 states, the action swap has a different effect under B's continuation than under Q's; per W4 cell the share is 22.9–30.5%. In the KV8 pool it is 15.3% [13.2, 17.3]; the W4-KV8 difference is descriptive. Strict sign reversals are rare: 3.2% of pooled W4 states, and 1.2–4.2% of states in each W4 and KV8 cell. They make up 8–14% of the off-diagonal mass. The rest is an effect under one controller and none under the other.

Why the mean interaction is near zero. $I = \Delta_Q - \Delta_B$ takes values in $\{-2, \ldots, +2\}$, so $E[I] = \sum_{k=-2}^{2} k\, P(I{=}k)$,

whereas the off-diagonal share is $P(I \neq 0)$. A mean near zero only says that states where the swap helps more under Q ($I > 0$) and states where it helps more under B ($I < 0$) are about equally frequent. In the W4 pool they are 13.6% and 13.4% of states, so the mean is +0.0 [−1.9, +2.0] pp although 26.9% of states depend on the controller. $E[I]$ is the difference between the action swap's average effect under the two controllers; the average value of a fixed replacement policy is given by $E[G]$, $E[H]$ and $E[F]$ instead. The off-diagonal share asks whether an action's consequence belongs to the action alone, and for about one in four W4 states it does not (new-task interval 22.1–32.0%). All 16 outcome patterns per cell are in Appendix C.

6 R3: Do Sampling and Environment Change the Answer?

Bank P skips text-only differences. We ask whether the picture changes at ordinary states, under another user simulator, at unseen tasks, and without a user simulator.

Ordinary states turn over less, mostly through text under our simulator. At ordinary states, replacing Q's action changes the outcome less often than at first divergences. Over blocks f01–f03 (task-equal; t intervals over block means), M per sub-bank is .078–.090 for the q4 W4 cells in U1 and U2B, .047–.090 for the q8 W4 cells and at most .047 for KV8. The net gain lies within ±.028 (U1, U2B) and ±.050 (U2Q), and every interval covers zero (Appendix E). These supersede the f01-only numbers, in which the q8 W4 cells looked lower (at most .047). Identical actions never change the outcome, as deter-

minism implies. After deduplication, 150 of the 208 outcome-changing cell×state rows (≈72.1%) are class (ii); the rows are not independent states, since the three blocks repeat the same tasks (Appendix Table 8): the same tool structure with different text, nearly always two natural-language replies. 26 involve a different tool name, call count or argument. This is a *composition share*: per instance, text, tool/argument and act/talk differences flip the outcome in 14.4%, 18.2% and 18.9% of rows. Text dominates because it is common (1045 of 1357 differing rows), not because each text difference matters more. Per sub-bank the text rate is .126–.182, and simple content features (a question, a confirmation request, a new entity on one side only) do not separate flipping from non-flipping text pairs. Bank P cannot see these differences by design. These flip labels come from one simulator; the sensitivity check below, run on f01 flips, found that they do not reproduce reliably under another.

Sensitivity to the user simulator. Most consequential differences at ordinary states are text, so the user simulator is a plausible mediator; changing it moves τ-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated class-(ii) flips of block f01 and 39 matched non-flipping states. Their BQ and QQ branches were each re-run three times with a different user simulator, Sonnet 5.5 answering through a file relay from Claude Code subagents. Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. Dropping two mechanically flagged replies leaves this unchanged (6, 6 and 25 of 37; Appendix Table 9). In either direction, 13 of the 39 original flip states flip again, against 11 of 39 new flips at matched non-flip states; this does not show that the two groups are equivalent. **The Bank U text-difference outcome labels did not reproduce reliably under this alternative relay setting.** This is not evidence that text consequences are simulator artefacts: the control changes the simulator model, adds sampling randomness and uses a relay interface whose responders may have seen other requests. Its scope is narrow: fixed histories from the original simulator, a stochastic relay rather than API Sonnet or real users, block f01 only and three replicates.

Unseen tasks: telecom holdout. Before any holdout unit was run, we froze a confirmatory protocol (sha256 `902892dd. . .`). It covers 34 unseen telecom tasks × 12 fresh nonce blocks × 3 cells (q4 GPTQ-W4, q8 GPTQ-W4, q8 AWQ-W4, weighted 1/2, 1/4, 1/4). The primary endpoint is G_{all} on a crossed 90% interval against ±3 pp; secondaries are two-sidedness and repeatability of task fragility (Appendix H). All 12 blocks were complete; two task-cell pairs were excluded as structural, and the complete-case analysis is identical. The primary endpoint is $G_{\text{all}} = -4.57$ pp, crossed 90% interval $[-7.17, -1.96]$ (block-t $[-6.08, -3.05]$). The pre-registered readings define only a positive gain, a gain above 3 pp and a small effect, so the decisional reading is *inconclusive*. Descriptively, the interval lies entirely below zero: on these unseen tasks, replacing the first divergent action with BF16's lowered average success. Both secondaries pass. Rescues and damages are each above zero ($r = +5.21$ $[+3.30, +7.11]$, $d = +9.77$ $[+6.45, +13.10]$ pp), and task fragility repeats across independent nonce halves ($\rho = 0.59$, permutation $p = .0001$). The holdout confirms what the protocol pre-specified, two-sided consequences and repeatable task fragility, and no positive average benefit of current-action replacement; it does not confirm the rest of the paper's development results.

No user simulator: BFCL and ALFWorld. BFCL multi-turn and ALFWorld have no simulated user, and each item runs once. Replacing a W4 agent's first divergent action changes the outcome in 5.2–24.6% of items, in both directions. The net gain covers zero except for BFCL q4 AWQ-W4, where it is -3.8 $[-6.5, -1.3]$ pp. Two-sided turnover is therefore not specific to a user simulator. Intervals here are item-level only.

7 R4: Can Beneficial Replacements Be Found in Advance?

G, H and F are oracle quantities that require running B and all four branches; a deployable rule sees only Q. Following Lotfi et al. (2026), we scored every Bank U f01 state at the token level (1140 cell × state rows, 58 flips), teacher-forcing both models on the generated ids of a_{B} and a_{Q}. We record truncated top-20 entropy, the top-1 mar-

gin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

Uncertainty locates divergence. Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8, stratified by cell and action length).

Among divergent states, it does not reliably identify consequences. Restricting to states where the actions differ removes the "any difference at all" signal. In U1 the effect vanishes: no Q-visible statistic has an interval excluding 0.5. Only q4 U2B keeps a moderate signal (mean entropy over a_{Q}, AUC 0.73 [0.53, 0.89]). At the first divergent token, entropy and KL do not separate flips from non-flips, with one exception on 4 flips (q4 U2Q, KL AUC 0.80 [0.65, 0.93]). No class of first divergent token reproduces the marker enrichment of Lotfi et al. (2026). The largest q4 enrichment is ordinary word choice (rate ratio 1.89, Fisher $p = .019$, not corrected). On development data, uncertainty thus marks *where* quantization changes the action much better than *which* changes matter. Whether a rule can still capture turnover, or net gain, is a separate question.

A frozen trigger, tested on held-out blocks. To test this, we froze a Q-visible trigger on f01 before any f02/f03 state was scored: escalate one decision when the minimum Q margin over a_{Q} is low, with per-cell thresholds at 5, 10 and 20% escalation rates (rule sha256 `335e737d`. . .). At the lower rates it mostly fires on actions that contain an exact top-2 tie, broken by a frozen hash. On held-out f02/f03 (1020 U1 rows, 69 turnover states) it escalated 50, 114 and 222 states. Net gains were $+4$, $+2$ and $+5$ (rescues/damages 6/2, 7/5, 16/11) against matched-count random expectations of $+0.21$, $+0.98$ and $+2.38$ (exact one-sided hypergeometric $p = .0345$, .417, .262). It captured 8, 12 and 27 turnover states against 3.48, 7.52 and 14.30 expected ($p = .0182$, .0613, .0002). No significance level was fixed in advance; at a nominal one-sided .05, the net-gain endpoint passes at the 5% rate only, and not after Holm correction over the three rates. The 20% threshold selected 222 of 1020 rows (21.8%); its net gain of about 0.49 pp exceeds the random expectation of about 0.23 pp by 0.26 pp ($p = .262$). The frozen Q-only margin rule enriched outcome-changing decisions in later nonce blocks, but has not been shown to select net-beneficial replacements reliably. This is a comparison with random selection on a fixed validation bank; two new nonce blocks do not establish routing gains on new tasks.

8 Discussion

When to escalate. Average success, the rate of action divergence and the gain from replacing an action answer different questions, and our data show they can disagree. The four W4 cells have small end-to-end differences with wide intervals (Appendix B). Their Bank P states replace actions with 15.5–20.9% turnover, but the pooled replacement gains are near zero. Reporting only the first two cannot tell whether a BF16 fallback is worth its cost. The outcome labels of text-only differences at ordinary states did not reproduce reliably under an alternative, stochastic relay simulator (§6). Text actions should be evaluated as actions, but their consequences should not be reported as a property of the action without a simulator control. We recommend reporting r, d and M next to any net replacement gain, each with its denominator. Fixed-pool and new-task intervals should be kept apart.

Which part to replace. The unit of replacement matters, and its best choice is not universal. In the q8 W4 cells, replacing the current action beats replacing the continuation. In q8 GPTQ-W4, handing over the rest of the episode adds further gain. In a severely damaged W3 cell, only the continuation matters (§5). These within-cell results suggest measuring G, H and F per deployment configuration rather than transferring one answer across quantizers.

Escalation can hurt. Replacements damage about as often as they rescue (§4). A handover can also remove a recovery the quantized agent would have made: in one frozen case (q4 AWQ-W4, retail task 61), Q recovers from its own failed write and succeeds, while B continuing after that write fails. In q4 KV8, continuing with B lowers success on average (§5), and on the unseen telecom tasks replacing the first divergent action lowers it by 4.6 pp (§6). A rule that escalates whenever B's action "looks better" would not gain.

What we do not claim. We do not claim success equivalence between BF16 and W4 agents; equivalence tests at ± 3 pp fail. G, H and F are oracle quantities, not the value of a deployable policy (§7). We do not claim a W$\times$KV interaction, and we do not present static precision restoration as a method (Appendix I).

Limitations

One model family, two sizes. All agents are Qwen3 models (4B and 8B). Effects may differ at other scales and in other families. End-to-end runs exist for larger Qwen models, but they have no crossed forks, and we draw no conclusion from them (Appendix D).

Simulated quantization. Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

One user simulator; a relay control. All τ^2-bench episodes use a single greedy simulator model, and continuation effects include its replies. The second-simulator control uses Sonnet 5.5 answering through a Claude Code subagent relay, which is stochastic and whose context we do not fully control.

LLM annotation. Process labels in Appendix G come from LLM annotators (Sonnet 5.5, two independent instances plus an adjudicator) and rule-based checks. No human annotation has been done.

Small confirmation and partial banks. The pre-registered holdout covers 34 telecom tasks and three cells. Its negative average gain is specific to telecom, these cells and this simulator, and its pre-specified readings did not include a negative-gain category. Bank P intervenes only at the first tool-structure divergence per task and block. Bank U samples one state per task and block and has three blocks, so its block intervals rest on two degrees of freedom. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

Ethics Statement

This work involves no crowdworkers and no personal data. All users are simulated by an LLM on public benchmark tasks, and all agents are openly released Qwen3 checkpoints. The process annotations were produced by LLMs on synthetic benchmark conversations. The main risk we see is over-reading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

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A Setup Details

Splits. The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from τ^2-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

Success endpoints. For retail and airline we use the final-database match with normal termination. This is not τ^2-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures. Infrastructure outages are missing, never failures, and a block with a missing unit leaves the primary analysis.

Fork mechanics. The forced action is returned by a fork agent for the one step that follows h_t. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. tau2 resets step and error counters when it starts from a history. The fork restores them to the values used at t, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount.

B End-to-End Success Comparison

The replacement gains G_{all}, G, H and F are not end-to-end comparisons. Table 3 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

Cell	Q / BF16 (%)	Q−BF16 [95% CI]	NI 3 pp	TOST ±3 pp
q4 GPTQ-W4	40.0 / 35.8	+4.2 [−2.4, +10.5]	pass	fail
q4 AWQ-W4	36.8 / 35.8	+0.9 [−5.8, +7.5]	fail	fail
q8 GPTQ-W4	30.1 / 30.4	−0.2 [−7.7, +6.8]	fail	fail
q8 AWQ-W4	31.5 / 30.4	+1.2 [−5.3, +7.6]	fail	fail
pool, all four		+1.5 [−2.7, +5.6]	pass	fail

Table 3: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence. Source: `analysis/insights/iter_4/out/success_ni_ALIGNED_20260930T213009Z.md`.

C All 16 Outcome Patterns and New-Task Intervals

Table 4 lists every four-branch pattern per cell; `0000` (39–58%) and 1111 (6–20%) dominate. Table 5 gives the new-task intervals for Table 2.

D Coverage Matrix

Table 6 lists the completed crossed-intervention coverage behind every result, from the W93 audit. It counts tasks, states and complete blocks after the strict admission rule; queued or running work is not counted.

E Bank U per Sub-Bank

Table 7 gives the per-state turnover behind §6, over blocks f01–f03; it supersedes the single-block f01 table of an earlier draft. Exact action agreement is 0.43–0.83 per cell and sub-bank.

Text content. Over f01–f03, an above-median relative length difference between the two texts co-occurs with outcome change in U1 and U2B of both models (flipping pairs among above- vs. below-median pairs: q4 U1 30/154 vs. 15/155; q8 U1 22/101 vs. 6/101; q4 U2B 33/138 vs. 11/139; q8 U2B 16/99 vs. 9/100). In f01 alone this was visible only for q4. Question, confirmation-request and new-entity flags show no consistent association. The features are regular expressions, and this is descriptive.

Shared fragility across quantizers (a lead). On the 255 shared q4 U2B states of f01–f03, cell pairs co-flip 21 times, against about 3.3 expected under independence, and 16 of the 21 co-flips go in the same direction. They come from 15 distinct states, 3 of which flip under all three cells. On q8 U2B, where every f01 flip was single-cell, 10 states now flip under two or more cells (14 pairwise co-flips against about 2.8 expected; 10 of 14 in the same direction). Joint flips almost never share an identical quantized action across cells (0 of 21 in q4, 2 of 14 in q8). We still read this as a lead toward *state-level* fragility, where different actions at the same state flip the outcome, and not yet as a finding: the co-flips are counted on one fixed set of states per block, with no correction.

A common controller (Q0). On the U2 subsample of f01–f03 continued by the common controller Q0, B's action leads to the identical outcome in every cell on all 146 shared states, which checks determinism. Turnover under Q0 is not consistently above or below the native turnover on the same rows (q4 AWQ-W4 U2B 8/58 vs. 5/58; q8 AWQ-W4 U2B 4/52 vs. 5/52; q4 KV8 U2B 1/58 vs. 2/58), so the f01 observation that Q0 turnover is at least the native one does not hold over three blocks. With 17–58 states per cell and sub-bank, this is too small to say whether the controller changes which actions matter.

F Concentration of Turnover in Tasks

This appendix completes the exploratory concentration result of §4 (development data, 85 tasks $\times$ 20 blocks). The per-task dispersion of M is 4.0–8.7 times its binomial null in the W4 and KV8 cells, and the top 20% of tasks hold 57–71% of all flips. We split the 20 blocks into two disjoint halves of 10 nonces. Task-level M correlates across halves (Spearman 0.52–0.86). Tasks that were net-rescue in one half have a mean net gain in the other half that is 14–51 pp higher than that of net-damage tasks, in all 12 cell-directions. A Freedman–Lane test conditions on measured baselines: QQ success propensity and BF16 success, plus trajectory length in a post-hoc variant. With them, one half's task rate still predicts the other's in all 6 configurations $\times$ 2 directions for both M and G (Holm-adjusted). This is a *retrospective conditional association* on viewed development data. It is carried by retail. Airline alone gives 8 of 12 for M (5 after Holm), and 3 of 12 under a heteroscedasticity-robust wild bootstrap. Direction persistence beyond flip opportunity holds more often for rescues than for damages, in 11 versus 6 of the 12 cell-directions. Different quantizers flip the same tasks: the task-identity null is rejected in 7 of 8 quantizer pairs. When both flip, however, they agree in direction only 42–81% of the time. This overlaps with the stable failure set of Jang et al. (2026). What is new here is the signed, two-sided view: shared fragility, but with near-random relative direction across quantizers.

G Controlled Injection and Process Audit

A natural explanation of R1 is a local-defect story: the quantized agent makes a local error, and the task fails unless the continuation recovers. Two analyses constrain this explanation. They limit what it can explain; they do not identify a mechanism.

$(Y_{BB}, Y_{BQ}, Y_{QB}, Y_{QQ})$	q4 AWQ	q4 GPTQ-W3	q4 GPTQ	q4 KV4r	q4 KV8	q8 AWQ	q8 GPTQ-W3	q8 GPTQ	q8 KV4r	q8 KV8	q8 RTN-W3
0000	634 (44.1%)	105 (45.3%)	472 (42.8%)	92 (39.5%)	286 (48.2%)	767 (49.1%)	130 (53.7%)	766 (52.0%)	113 (47.3%)	497 (58.1%)	141 (55.3%)
0001	55 (3.8%)	7 (3.0%)	29 (2.6%)	3 (1.3%)	14 (2.4%)	66 (4.2%)	2 (0.8%)	36 (2.4%)	8 (3.3%)	15 (1.8%)	3 (1.2%)
0010	57 (4.0%)	13 (5.6%)	54 (4.9%)	13 (5.6%)	11 (1.9%)	38 (2.4%)	8 (3.3%)	34 (2.3%)	10 (4.2%)	15 (1.8%)	20 (7.8%)
0011	58 (4.0%)	6 (2.6%)	71 (6.4%)	4 (1.7%)	37 (6.2%)	30 (1.9%)	9 (3.7%)	49 (3.3%)	3 (1.3%)	39 (4.6%)	0 (0.0%)
0100	50 (3.5%)	6 (2.6%)	36 (3.3%)	9 (3.9%)	8 (1.3%)	53 (3.4%)	8 (3.3%)	45 (3.1%)	13 (5.4%)	20 (2.3%)	1 (0.4%)
0101	21 (1.5%)	4 (1.7%)	17 (1.5%)	4 (1.7%)	11 (1.9%)	36 (2.3%)	7 (2.9%)	12 (0.8%)	5 (2.1%)	5 (0.6%)	2 (0.8%)
0110	22 (1.5%)	5 (2.2%)	14 (1.3%)	2 (0.9%)	2 (0.3%)	33 (2.1%)	4 (1.7%)	16 (1.1%)	2 (0.8%)	5 (0.6%)	1 (0.4%)
0111	41 (2.8%)	5 (2.2%)	21 (1.9%)	7 (3.0%)	13 (2.2%)	51 (3.3%)	6 (2.5%)	31 (2.1%)	7 (2.9%)	7 (0.8%)	6 (2.4%)
1000	51 (3.5%)	8 (3.4%)	32 (2.9%)	19 (8.2%)	16 (2.7%)	67 (4.3%)	10 (4.1%)	75 (5.1%)	16 (6.7%)	28 (3.3%)	32 (12.5%)
1001	23 (1.6%)	4 (1.7%)	15 (1.4%)	2 (0.9%)	5 (0.8%)	33 (2.1%)	7 (2.9%)	24 (1.6%)	9 (3.8%)	5 (0.6%)	0 (0.0%)
1010	18 (1.3%)	5 (2.2%)	13 (1.2%)	9 (3.9%)	4 (0.7%)	17 (1.1%)	5 (2.1%)	17 (1.2%)	3 (1.3%)	3 (0.4%)	20 (7.8%)
1011	36 (2.5%)	9 (3.9%)	33 (3.0%)	9 (3.9%)	8 (1.3%)	32 (2.0%)	3 (1.2%)	19 (1.3%)	2 (0.8%)	11 (1.3%)	6 (2.4%)
1100	44 (3.1%)	3 (1.3%)	51 (4.6%)	9 (3.9%)	41 (6.9%)	43 (2.8%)	10 (4.1%)	82 (5.6%)	9 (3.8%)	46 (5.4%)	2 (0.8%)
1101	36 (2.5%)	7 (3.0%)	27 (2.5%)	3 (1.3%)	9 (1.5%)	38 (2.4%)	4 (1.7%)	23 (1.6%)	1 (0.4%)	11 (1.3%)	1 (0.4%)
1110	51 (3.5%)	9 (3.9%)	15 (1.4%)	10 (4.3%)	7 (1.2%)	65 (4.2%)	4 (1.7%)	34 (2.3%)	6 (2.5%)	10 (1.2%)	4 (1.6%)
1111	242 (16.8%)	36 (15.5%)	202 (18.3%)	38 (16.3%)	121 (20.4%)	193 (12.4%)	25 (10.3%)	210 (14.3%)	32 (13.4%)	138 (16.1%)	16 (6.3%)
n	1439	232	1102	233	593	1562	242	1473	239	855	255

Table 4: All 16 four-branch outcome patterns per cell: count (share of the cell's Bank P states). Pattern digits are $(Y_{BB}, Y_{BQ}, Y_{QB}, Y_{QQ})$. KV4r: Q/K-rotated KV4; W3 and KV4r cells are stress cells with 3 blocks. Source: `analysis/fork/out/ghf_20261001T211519Z.md` (§3).

Cell	G	H	F	$G-H$	$F-G$	$F-H$
q4 GPTQ-W4	$[-10.3, +4.2]$	$[-4.0, +6.0]$	$[-9.9, +4.7]$	$[-11.2, +3.6]$	$[-4.7, +6.1]$	$[-11.1, +4.4]$
q4 AWQ-W4	$[-6.5, +6.1]$	$[-3.7, +5.5]$	$[-7.4, +5.8]$	$[-7.7, +5.1]$	$[-6.2, +5.4]$	$[-8.7, +4.7]$
q8 GPTQ-W4	$[-3.3, +10.5]$	$[-3.8, +4.9]$	$[-2.0, +13.0]$	$[-3.2, +9.3]$	$[-3.0, +7.1]$	$[-2.0, +12.5]$
q8 AWQ-W4	$[-4.1, +8.7]$	$[-7.1, +4.6]$	$[-6.2, +7.3]$	$[-1.6, +8.6]$	$[-7.6, +4.2]$	$[-3.8, +7.5]$
W4 pool	$[-2.9, +3.8]$	$[-2.7, +2.8]$	$[-3.1, +4.4]$	$[-3.1, +3.7]$	$[-2.8, +3.1]$	$[-3.5, +4.5]$
q4 KV8	$[-9.4, +7.3]$	$[-6.4, +1.1]$	$[-9.3, +7.7]$	$[-7.1, +9.7]$	$[-5.7, +5.3]$	$[-6.7, +9.5]$
q8 KV8	$[-4.0, +6.9]$	$[-3.5, +2.9]$	$[-3.5, +8.1]$	$[-3.8, +7.2]$	$[-2.5, +5.0]$	$[-3.1, +8.7]$
KV8 pool	$[-5.3, +5.6]$	$[-3.7, +0.9]$	$[-4.5, +5.8]$	$[-3.8, +6.6]$	$[-2.8, +3.9]$	$[-3.0, +7.3]$
q4 GPTQ-W3^s	$[-9.7, +6.6]$	$[-6.6, +14.5]$	$[-6.8, +9.4]$	$[-15.0, +4.0]$	$[-5.2, +10.5]$	$[-12.2, +6.6]$
q4 KV4-rot^s	$[-3.4, +14.2]$	$[+2.1, +17.0]$	$[+3.0, +21.6]$	$[-13.5, +7.0]$	$[-1.3, +17.1]$	$[-6.2, +12.2]$
q8 GPTQ-W3^s	$[-6.7, +12.1]$	$[-7.2, +8.1]$	$[-6.6, +11.8]$	$[-7.7, +11.2]$	$[-9.1, +9.1]$	$[-6.8, +11.0]$
q8 KV4-rot^s	$[-6.6, +13.8]$	$[-11.2, +8.5]$	$[-6.6, +15.9]$	$[-3.3, +12.2]$	$[-7.2, +9.9]$	$[-2.5, +14.6]$
q8 RTN-W3^s	$[-4.7, +3.9]$	$[+5.9, +25.9]$	$[+9.4, +28.2]$	$[-24.7, -7.1]$	$[+10.2, +27.8]$	$[-7.1, +13.3]$

Table 5: New-task generalisation: crossed task×block bootstrap 95% intervals for the estimates of Table 2 (pp; same Bank P states). These answer whether the effect would hold for new tasks drawn like the panel; the fixed-pool intervals of Table 2 treat the 85 tasks as fixed. ^{s}stress cell, 3 blocks. Source: `analysis/fork/out/ghf_20261001T211519Z.md` (§1).

Controlled defect injection. We froze 30 ordinary states from q4 GPTQ-W4 trajectories, all from one development block. At each state we forced one of three actions as the next step: an INValid action, a matched VALid action, or a process-perturbation control (PP) that repeats the last successful read. Either B or Q then continued. The defect categories were wrong_id, wrong_tool and premature_write, with 10 states each. Every state passed a replay audit of 22 recorded fields (VAL×Q against the original episode). Three fields were not recorded: the counters right after the restore, the rendered-prompt hash sequence and the tool-schema hash sequence.

Result: two-sided turnover, no clear error-specific difference detected. Injected defects change the outcome in 7/30 states under B (4 rescues, 3 damages) and 4/30 under Q (2 and 2). The process-perturbation control changes 5/30 under both controllers (Table 10). The error-specific effect, INV minus PP, is 0.00 $[-0.13, +0.13]$ under B and -0.03 $[-0.20, +0.10]$ under Q. The controller dependence D is nonzero in 8 of 30 states, but its mean is $+0.03$ $[-0.17, +0.27]$. So injected defects cause heterogeneous two-sided turnover, and at $n = 30$ states we detect no clear difference from a generic process perturbation and no mean controller difference. The intervals are wide, so this does not show that errors and generic perturbations are equivalent. An earlier reading of the same data, that injected defects "seldom change outcomes", was wrong and has been retracted (Appendix I).

Manipulation checks limit the categories. Two of 30 states failed a manipulation check: an injected write was rejected, and a valid reference itself returned "not found". Restricting to the 28 passing states leaves the reading unchanged. The wrong_tool category mostly produced argument errors: 8 of 10 injected calls failed with an unexpected-keyword error. We therefore call it a *tool-call/schema error perturbation*,

Cell	τ^2-bench retail (Bank P)	τ^2-bench airline (Bank P)	telecom dev (Bank P)	Bank U (retail / airline)	BFCL / ALFWorld
q4 GPTQ-W4	58 t · 727 st · 20/20 b	27 t · 375 st · 20/20 b	holdout dirs exist (unread)	3 b · 393 / 177 rows	788 items · 573 st / –
q4 AWQ-W4	58 t · 986 st · 20/20 b	27 t · 453 st · 20/20 b	–	3 b · 393 / 177 rows	788 items · 642 st / –
q4 KV8	58 t · 412 st · 22/22 b	27 t · 181 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q8 GPTQ-W4	58 t · 1017 st · 20/20 b	27 t · 456 st · 20/20 b	80 t · 477 st · 6/6 b; holdout dirs	3 b · 393 / 177 rows	788 · 533 st / 134 · 115 st
q8 AWQ-W4	58 t · 1087 st · 20/20 b	27 t · 475 st · 20/20 b	80 t · 480 st · 6/6 b; holdout dirs	3 b · 393 / 177 rows	788 · 548 st / 134 · 119 st
q8 KV8	58 t · 585 st · 22/22 b	27 t · 270 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q4 GPTQ-W3	58 t · 162 st · 3/3 b	27 t · 70 st · 3/3 b	–	–	788 · 720 st / –
q4 KV4-rot	58 t · 164 st · 3/3 b	27 t · 69 st · 3/3 b	–	–	– / –
q4 RTN-W3	0 t · 0 st · 0/3 b	0 t · 0 st · 0/3 b	–	–	– / –
q8 GPTQ-W3	58 t · 166 st · 3/3 b	27 t · 76 st · 3/3 b	–	–	788 · 696 st / 134 · 130 st
q8 KV4-rot	58 t · 165 st · 3/3 b	27 t · 74 st · 3/3 b	–	–	– / –
q8 RTN-W3	58 t · 174 st · 3/3 b	27 t · 81 st · 3/3 b	–	–	– / –
q4 / q8 KV4 (unrotated)	–	–	–	–	788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st

Table 6: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (q8 KV8 f03 finished after the coverage snapshot; Appendix E). "holdout dirs": telecom holdout fork directories exist and were not opened. The q14, q32 and l8 checkpoints of the source matrix have end-to-end runs only, and q4 RTN-W3 has no complete Bank P block. The telecom development analysis of §6 uses the planning fit over these 6 blocks. Source: `analysis/fork/out/ghf_20261001T211519Z.md` (§5). The end-to-end q4 AWQ-W3 run is incomplete: two airline episodes hit 600 s serving timeouts (infrastructure), so it is not reported; its completed episodes and failure logs are kept.

Cell	U1 (n=255)	U2B (n=255)	U2Q (n=60)
q4 GPTQ-W4	.090 (11/12)	.082 (10/11)	.083 (4/1)*
q4 AWQ-W4	.082 (7/14)	.078 (7/13)	.083 (3/2)
q4 KV8	.047 (4/8)	.039 (5/5)	.017 (0/1)
q8 GPTQ-W4	.047 (8/4)	.067 (9/8)	.050 (3/0)*
q8 AWQ-W4	.086 (14/8)	.090 (15/8)	.067 (3/1)
q8 KV8	.035 (5/4)	.031 (6/2)	0 (0/0)

Table 7: Bank U, blocks f01–f03: per-state turnover M (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). G lies within ± 0.028 in U1 and U2B and within ± 0.050 in U2Q, and every interval includes 0. Intervals are t intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]); all are in the source. *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals. Source: `analysis/fork/out/bank_u_f01_f02_f03_20261002T003037Z.md`.

Model:sub	(i) identical	(ii) same tool structure, text	(iii) different tool or args	(iv) act vs. talk
q4:U1	399: 0	309: 45 (.146)	19: 1 (.053)	38: 10 (.263)
q8:U1	483: 0	202: 28 (.139)	47: 11 (.234)	33: 4 (.121)
q4:U2B	430: 0	277: 44 (.159)	24: 4 (.167)	34: 3 (.088)
q8:U2B	470: 0	199: 25 (.126)	40: 9 (.225)	56: 14 (.250)
q4:U2Q	108: 0	63: 10 (.159)*	5: 0	4: 1 (.250)
q8:U2Q	129: 0	33: 6 (.182)*	11: 1 (.091)	7: 0
unique rows	1943: 0	1045: 150 (.144)	143: 26 (.182)	169: 32 (.189)

Table 8: Which kind of action difference changes the outcome at ordinary states (Bank U, blocks f01–f03). Entries: "state rows: outcome-changing rows (flip rate)", pooled over the three cells of a model within a sub-bank. The last row counts each (cell, state) once: q4/q8 GPTQ-W4's U2Q states repeat their U1 states (* includes 5 and 3 repeated class-(ii) flips), so 216 flipping rows become 208. In block f01 alone the corresponding counts were 39 of 56 flips in class (ii), with rates .126/.109/.200 for (ii)/(iii)/(iv). Source: `analysis/fork/out/bank_u_f01_f02_f03_20261002T003037Z.md`; f01: `analysis/fork/out/bank_u_f01_20261001T134456Z.md`.

not a wrong-tool test. premature_write is mostly failure→failure: the valid arm under Q already fails in 7/10 of these states, and turnover is 0/10 under both controllers. The event chain error → correction → related requirement satisfied completes in only 6 (B) and 7 (Q) of 29 injected errors, although 17 are corrected under each controller. Injections add tool errors ($+0.43$ under B, $+0.57$ under Q per branch).

Replication on q8 GPTQ-W4. The same design on 30 frozen q8 states (block q3 of the GPTQ-W4 run; 10 per category) gives the same reading (Table 11). All 30 states pass the replay audit. Injected defects change the outcome in 5/30 states under each controller (3 rescues, 2 damages), the process-perturbation control in 6/30 (B) and 5/30 (Q). The error-specific effect is -0.03 [-0.13, $+0.07$] under B and -0.13 [-0.30, $+0.03$] under Q; the latter interval still allows a sizeable error-specific harm, and D is nonzero in 6 of 30 states with mean 0.00 [-0.17, $+0.17$]. Under Q, the process-perturbation control itself rescues 5 states and damages none ($+0.17$ [$+0.03$, $+0.30$], McNemar $p = 0.0625$). Manipulation checks fail more often than in q4: 7 of the 10 premature writes failed the check (6 were rejected by the environment, 1 executed without changing the database), and 2 valid references returned "not found". On the 21 passing states, INV vs VAL turnover is 5 (B) and 4 (Q), and $D \neq 0$ in 5. The q8 prema-

Analysis	flips	retained	rev.	vanished (s/f/tie)	ctrl	new flips (resc./dam.)
All replies	39	7 [0.07, 0.34]	6	26 (15/2/9)	39	11 [0.15, 0.45] (7/4)
Flagged dropped	37	6 [0.06, 0.32]	6	25 (15/2/8)	39	11 [0.15, 0.45] (7/4)

Table 9: Bank U simulator control (block f01; Sonnet 5.5 relay user, $R = 3$). Fates use the sign of the replicate-mean flip indicator; s/f: both arms succeed/fail. "Flagged dropped": the two arm replicates with a mechanically flagged reply are set to missing. Replicates of one arm disagree in 16/39 (BQ) and 16/39 (QQ) flip states and in 10/39 and 8/39 control states. Per single replicate, retained flips are 5, 7 and 4 of 39. Source: `analysis/fork/out/simctrl_20261001T212742Z.md`, `analysis/fork/out/simctrl_relay_flagged_20261001T225847Z.md`.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	4/3	7/30	$+0.03$ $[-0.13, +0.20]$
B	PP vs VAL	3/2	5/30	$+0.03$ $[-0.10, +0.20]$
B	INV vs PP	2/2	4/30	0.00 $[-0.13, +0.13]$
Q	INV vs VAL	2/2	4/30	0.00 $[-0.13, +0.13]$
Q	PP vs VAL	3/2	5/30	$+0.03$ $[-0.10, +0.17]$
Q	INV vs PP	2/3	5/30	-0.03 $[-0.20, +0.10]$
Controller dependence D ($\neq 0$ in 8/30 states)				$+0.03$ $[-0.17, +0.27]$

Table 10: Defect injection on 30 frozen q4 states (development, one block): per-state rescue/damage of one arm against another, under the BF16 (B) or quantized (Q) continuation controller. $D = [Y(\text{INV}, B) - Y(\text{VAL}, B)] - [Y(\text{INV}, Q) - Y(\text{VAL}, Q)]$. Task-clustered bootstrap intervals; all exact McNemar $p \geq 0.25$. Source: `analysis/recovery/inject_q4/summary_v2.md`.

ture_write category therefore mostly did not inject a write.

Process audit (LLM-annotated). We sampled 80 development fork states, 20 per outcome sign. Two independent Sonnet 5.5 instances annotated them blind, a third instance adjudicated, and rule-based programmatic labels were added. These annotations are a *process description* conditional on outcome-defined groups. They are not causal evidence, not prevalences and not human labels. Three patterns stand out (Table 12). First, local defects and final failure do not correspond one-to-one. In damage states, where Q's action led to success and B's to failure, Q's first-action defect is corrected later in 10 of 20 QQ branches. Second, whether a write executes does not separate the failing from the succeeding continuation of the same state (rescue 17 vs. 17; damage 15 vs. 14). Third, the failing continuation more often carries the rubric label `unintended_or_unconsented_change` (rescue: QQ 9 vs. BQ 2, paired difference $+0.35$ $[+0.08, +0.57]$; damage: 1 vs. 9, -0.40 $[-0.62, -0.11]$). In damage states, 7 of these 20 BQ branches carry the label on writes *with* recorded consent. That is the semantic "wrong change" part of the label, and it overlaps with the database-defined outcome by construction. Loops and budget exhaustion also mark failing branches (rescue: QQ 5 vs. BQ 0). We make no claim about consent. The programmatic consent field agrees poorly with the annotators, so we use it only as a mechanical row. Re-labelling 30 states with fresh instances of the same model family reproduces the key pattern. That shows reproducibility, not correctness.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	3/2	5/30	$+0.03$ $[-0.10, +0.17]$
B	PP vs VAL	4/2	6/30	$+0.07$ $[-0.07, +0.23]$
B	INV vs PP	1/2	3/30	-0.03 $[-0.13, +0.07]$
Q	INV vs VAL	3/2	5/30	$+0.03$ $[-0.10, +0.17]$
Q	PP vs VAL	5/0	5/30	$+0.17$ $[+0.03, +0.30]$
Q	INV vs PP	2/6	8/30	-0.13 $[-0.30, +0.03]$
Controller dependence D ($\neq 0$ in 6/30 states)				0.00 $[-0.17, +0.17]$

Table 11: Defect injection on 30 frozen q8 GPTQ-W4 states (development, block q3), as in Table 10. Task-clustered bootstrap intervals; exact McNemar $p \geq 0.0625$. Source: `analysis/recovery/inject_q8/summary_v2.md`.

G.1 Process audit (Table 4 v2)

The core "local defect" combines groundedness, argument validity and permission/consent judgements. The tool-validity field (inter-annotator $\kappa = 0.29$) is excluded from it. Mechanical fields agree with the programmatic verifier (write executed vs. the rubric's irreversible-change field, $\kappa = 0.99$). The consent fields agree only moderately ($\kappa = 0.44$ on writes both see).

H Holdout Pre-Registration

The protocol `docs/insights/PREREG_FORK_CONFIRM.md` was frozen before any holdout unit was run. Its sha256 is `902892dd63e99e16ab9beefee9d39319d835eacdada3d4d92c4a1022d13fd622`. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- halves $H_A = \{h01, h04, h05, h09, h11, h12\}$ and H_B, the rest of $h01$–$h12$;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);

Label (source)	Comparison	rescue	damage	both succeed	both fail
unintended_or_unconsented_change (LLM)	QQ vs BQ	9 vs 2, +0.35 [+0.08, +0.57]	1 vs 9, −0.40 [−0.62, −0.11]	1 vs 1	5 vs 7, −0.10 [−0.35, +0.17]
write executed (programmatic)	QQ vs BQ	17 vs 17, 0.00 [−0.25, +0.25]	15 vs 14, +0.05 [−0.26, +0.35]	10 vs 9	11 vs 13
executed write w/o recorded consent (prog.)	QQ vs BQ	9 vs 5, +0.20 [−0.06, +0.43]	3 vs 3, 0.00 [−0.21, +0.21]	3 vs 2	4 vs 3
loop or budget exhaustion (LLM)	QQ vs BQ	5 vs 0, +0.25 [+0.04, +0.47]	0 vs 2, −0.10 [−0.30, +0.08]	1 vs 2	7 vs 4
action judged indefensible (LLM)	Q vs B action	7 vs 8, −0.05 [−0.37, +0.28]	12 vs 5, +0.35 [−0.05, +0.65]	10 vs 6	12 vs 9
local defect, core (LLM)	Q vs B action	4 vs 6, −0.10 [−0.33, +0.14]	12 vs 7, +0.25 [−0.09, +0.53]	6 vs 3	11 vs 7
QQ first-action defect corrected later (LLM)	single branch	4/20 [0.08, 0.42]	10/20 [0.30, 0.70]	4/20	7/20

Table 12: Process audit of 80 development fork states, 20 per outcome sign (rescue: $Y_{\mathrm{BQ}} = 1, Y_{\mathrm{QQ}} = 0$; damage: $Y_{\mathrm{BQ}} = 0, Y_{\mathrm{QQ}} = 1$). **LLM-annotated** by Sonnet 5.5 (two independent blind instances and an adjudicating instance), plus rule-based programmatic labels; no human labels. Counts are first vs. second of $n = 20$ states. Intervals are within-state paired differences with Newcombe 95% CIs; single-branch rows use Wilson 95% intervals. This is a process description conditional on outcome-defined groups: not causal, not prevalence. The label name is the rubric category; we make no claim about consent. Source: `analysis/annotation/wp5/table4_v2.md`.

- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside ± 3 pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);
- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Table 13 gives the confirmatory result. Strata (six families, Holm within family) are descriptive here. Every level estimate is ≤ 0, and the Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues (-9.09 pp) and the fewest-faults tercile (-6.37 pp).

Cell (weight)	units	excl.	G	r	d	ρ_c
q4 GPTQ-W4 (1/2)	396	1	−3.03 [−7.86, +1.80]	+5.05	+8.08	0.668
q8 GPTQ-W4 (1/4)	408	0	−5.64 [−9.33, −1.94]	+4.41	+10.05	0.374
q8 AWQ-W4 (1/4)	396	1	−6.57 [−11.98, −1.15]	+6.31	+12.88	0.644
Weighted (G_{all})			−4.57 [−7.17, −1.96]	+5.21	+9.77	0.588

Table 13: Telecom holdout, pre-registered analysis (34 tasks × 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional (G_{all}, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search). Source: `analysis/fork/out/holdout_confirm_20261002T012353Z.md`.

I Withdrawn and Negative Results

W×KV interaction (not supported). An earlier q4 analysis reported a W4×KV4 interaction of -14.9 [-23.8, -6.0] pp ($k = 3$). Its KV16 reference lacked the Q/K rotation used for KV4, so rotation and quantization were confounded. With a matched rotated reference, the q4 interaction is -9.0 [-30.6, $+12.5$]. The pre-registered q8 replication gives $+2.1$ [-5.8, $+10.1$] over 6 blocks, crossed [-8.2, $+12.4$]. The replication is not supported, and we make no W×KV claim.

Static precision restoration (appendix only). We tested whether restoring selected modules to higher precision recovers a damaged RTN-W3 model. We report this line as a mechanism stress test, not a method. In RTN-W3 the gains largely tracked the number of restored bits. A random MLP arm also recovered much of the gap, and no arm beat whole-model GPTQ-W4 at matched storage. [TODO RESTORE: the q8 RTN-W3 attention vs. byte-matched random-attention result (one seed) is cited only after its restoration output is checked; it is not used for any claim]

Success equivalence. Not established (Table 3).

"Errors are absorbed". An interim reading of the injection study said that injected defects seldom change outcomes. The per-state analysis shows two-sided turnover (Appendix G), so this reading is retracted.

Superseded estimand names. The W89 draft reported $A = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$, $C = Y_{\mathrm{BB}} - Y_{\mathrm{BQ}}$ and I. In the present notation $A = \Delta_{\mathrm{B}} = F - H$ and $C = F - G$; the values are unchanged (Table 2).